

Count-Only Inverse-Variance Weighting for Intervention Harvesting

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Abstract

Intervention-harvesting estimators of position bias [1] combine per-cell click-rate estimates across many $(q, d, \text{position})$ cells whose impression counts span several orders of magnitude. Under the Position-Based Model (PBM), with cell-level relevance treated as homoscedastic, the *harmonic* weight $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ is the count-only inverse-variance choice—Aitken’s classical generalised-least-squares solution [2] restricted to count-only weights—and an unconditional Cauchy–Schwarz inequality shows it minimises the delta-method variance *surrogate* $V_k + V_{k'}$ for any impression counts. We propose the plug-in delta-method asymptotic variance $\tilde{V}(\omega)$ of the self-normalised weighted ratio estimator [9, 11] as a finite-sample diagnostic that is well aligned with the conditional-on-counts efficiency question, in contrast to a naive pair-bootstrap. On the full KDD Cup 2012 Track 2 sponsored-search log (235M impressions, 4.6M interventional pairs), the plug-in asymptotic relative efficiency of harmonic over uniform is 4.53, well-calibrated against click-resample standard error (1–2% at two sample fractions), and robust to weight-capping (ARE ≥ 4.1 at the 99.99th percentile). PBM is a known approximation; a Cochran- Q diagnostic on high-information cells quantifies its mis-specification at $Q/\text{df} = 3.83$. Under such mis-specification different weighting schemes target different weighted averages of cell-specific ratios—an estimand caveat we discuss explicitly. Synthetic (43%–54% MSE reduction at high imbalance) and Yahoo-LTR semi-synthetic ($\sim 40\%$ variance reduction at $r = 50$) experiments with ground truth corroborate the efficiency prediction.

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1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [6]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k/p_{k'}$ from naturally occurring ranking variations without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions $(1, 2)$), **60.7% of pairs**

have $N_1 + N_2 \leq 5$ while 0.1% of pairs carry 42% of all impressions [7]. Public benchmarks hide this shape: the released Baidu-ULTR [14] aggregation has 94% of (q, d, pos) cells with $N = 1$ and 78% of adjacent-pair impression ratios within $[0.5, 2]$ *because* the publication pipeline deduplicates impressions per cell—an artefact of release engineering, not of the underlying production logs. The unaggregated KDD release preserves the heavy-tailed production distribution.

Equal-weighting heteroscedastic estimators is statistically sub-optimal by a century of classical theory: Aitken’s generalised-least-squares theorem [2] mandates inverse-variance weights, a result the meta-analysis literature [3, 5] has championed for fifty years. Recent ULTR work [1, 12] introduced *harmonic* weighting $\omega_i = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ to reduce variance versus uniform on heavy-tailed counts, but existing validation has relied on *bootstrap* resampling – which we show is provably the wrong instrument for inverse-variance-style weights on singleton-dominated data (Section 3).

Contributions. (1) We identify harmonic as the count-only inverse-variance choice under the standard PBM with homoscedastic relevance (Theorem 2), prove an unconditional Cauchy–Schwarz reduction of the delta-method variance *surrogate* $V_k + V_{k'}$ (Theorem 3, conditional on the surrogate objective and unconditional in the counts), and characterise the exact asymmetric surrogate oracle of which harmonic is the $\alpha = \beta$ special case (Theorem 4). **(2)** We adopt the plug-in delta-method asymptotic variance $\tilde{V}(\omega)$ as a finite-sample efficiency diagnostic aligned with the conditional-on-counts question, and use empirical click-resample SE as its calibration baseline; we discuss why pair-bootstrap is misaligned with this question on heavy-tailed data. **(3)** On the full KDD log we measure $\text{ARE}(\text{harmonic}, \text{uniform}) = 4.53$, calibrated within 1–2% across two sample fractions, and robust to weight-capping at the 99.99th percentile (ARE ≥ 4.1). We quantify PBM mis-specification via Cochran’s $Q/\text{df} = 3.83$ and discuss the estimand caveat that follows. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

2 Estimator and Theory

Under PBM, $P(\text{click} \mid q, d, k) = p_k r_{q,d}$. Given an interventional set $I_{k,k'}$ of (q, d) pairs that appeared at both positions k and k' , the position-bias ratio is estimated as [1]

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \hat{\theta}_k^i}{\sum_i \omega_i \hat{\theta}_{k'}^i}, \quad \hat{\theta}_k^i = \frac{C_k^i}{N_k^i}. \quad (1)$$

This is a self-normalised weighted ratio estimator. We say a weight scheme is *ratio-unbiased* when the ratio of conditional expectations of numerator and denominator equals $p_k/p_{k'}$, i.e. $\mathbb{E}[A \mid \mathbf{N}] / \mathbb{E}[B \mid \mathbf{N}] = p_k/p_{k'}$ (*not* $\mathbb{E}[\hat{R}] = p_k/p_{k'}$; the finite-sample bias of the

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self-normalised ratio is $O(1/N)$ under standard regularity). Under PBM, any count-only weight $\omega_i = \omega(N_k^i, N_{k'}^i)$ preserves ratio-unbiasedness in this sense [1].

PROPOSITION 1 (CONDITIONAL DELTA-METHOD VARIANCE). *By the delta method [11], $\text{Var}(\hat{R} \mid \mathbf{N}) \approx R^2[\alpha V_k(\omega) + \beta V_{k'}(\omega)]$ with $V_r(\omega) = \sum_i \omega_i^2 / N_r^i / (\sum_i \omega_i)^2$, $\alpha = (1 - p_k)/p_k$, $\beta = (1 - p_{k'})/p_{k'}$.*

Harmonic as count-only IVW. For Bernoulli observations, $\text{Var}(\hat{\theta}_k^i) = \theta_k^i(1 - \theta_k^i)/N_k^i$. Under PBM, $\theta_k^i = p_k r_i$, so the Bernoulli variance contains a cell-level relevance r_i that is generally heterogeneous. After suppressing this cell-level relevance heterogeneity, the inverse-variance optimum of Aitken [2] restricted to count-only weights is the harmonic mean of the two cell counts. We make this precise as a statement about the variance *surrogate* obtained by ignoring cell-level relevance variation:

THEOREM 2 (COUNT-ONLY IVW ON THE SURROGATE). *Among all non-negative weights $\omega_i = \omega(N_k^i, N_{k'}^i)$, harmonic weighting minimises the count-only delta-method variance surrogate $V_k(\omega) + V_{k'}(\omega)$.*

THEOREM 3 (SURROGATE GUARANTEE, UNCONDITIONAL IN COUNTS). *For any impression counts $\{N_k^i, N_{k'}^i\}_{i=1}^n$, $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(\mathbf{1}) + V_{k'}(\mathbf{1})$. The guarantee is unconditional in the counts but conditional on $V_k + V_{k'}$ as the objective; it implies improved $\text{Var}(\hat{R} \mid \mathbf{N})$ under PBM when $\alpha \approx \beta$ (Proposition 1 below).*

SKETCH. Substituting $\omega_i^h = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ collapses the LHS to $1/W$ with $W = \sum_i \omega_i^h$. The RHS is $(1/n^2) \sum_i 1/\omega_i^h$. Cauchy–Schwarz gives $n^2 \leq W \sum_i 1/\omega_i^h$. $\square$

THEOREM 4 (EXACT ASYMMETRIC ORACLE). $\alpha V_k(\omega) + \beta V_{k'}(\omega) \geq (\sum_i 1/s_i)^{-1}$ with $s_i = \alpha/N_k^i + \beta/N_{k'}^i$, attained at $\omega_i^* \propto N_k^i N_{k'}^i / (\alpha N_{k'}^i + \beta N_k^i)$. Harmonic is the $\alpha = \beta$ special case.

Theorem 3 is unconditional in the counts – a property that distinguishes harmonic from the min-count heuristic of [1], which has no such guarantee. Theorem 4 provides the oracle benchmark; in synthetic experiments substituting the true α, β into ω^* yields variance within $\pm 2\%$ of harmonic.

3 Plug-in Asymptotic Variance: the Right Instrument

$\tilde{V}$ in closed form. With $A = \sum_i \omega_i \theta_k^i$, $B = \sum_i \omega_i \theta_{k'}^i$, and clicks at k and k' conditionally independent given counts, the delta method gives

$$\tilde{V}(\omega) = \frac{\sum_i \omega_i^2 \theta_k^i (1 - \theta_k^i) / N_k^i}{A^2} + \frac{\sum_i \omega_i^2 \theta_{k'}^i (1 - \theta_{k'}^i) / N_{k'}^i}{B^2}, \quad (2)$$

$\text{SE}(\hat{R}) = |\hat{R}| \sqrt{\tilde{V}(\omega)}$. Plugging in $\hat{\theta}_i$ (with light shrinkage toward the marginal) yields the textbook self-normalised IS variance estimator [8–10].

ARE. For schemes a, b , $\text{ARE}(a, b) = \tilde{V}(\omega_b) / \tilde{V}(\omega_a)$: scheme b needs $\text{ARE}(a, b)$ times the data of a for equal precision. $\tilde{V}$ scales as $1/\text{ESS}(\omega)$, with Kong’s effective sample size [8] $\text{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$. free optimum.

Table 1: Plug-in delta-method asymptotic variance $\tilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\text{pairs}} = 4.61\text{M}$). Harmonic, min, and the count-only oracle all reach ARE ≈ 4.5 . The full- θ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
Min [1]	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	4.53	1,849
Count-oracle (0.5,1)	1.715	2.02e-3	4.57	1,760
θ -aware oracle	2.343	0.148	0.002	1.1

Table 2: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ quartile on full KDD. Schemes are tied on the noise-dominated quartiles; harmonic dominates $\sim 5\times$ on the high-information quartile where the actual estimation signal lives.

$N_1 + N_2$ quartile	n pairs	$\tilde{V}_U$	ARE(h, u)
Lowest	1.99M	2.3e-5	1.02
Middle-low	1.38M	1.9e-5	1.09
Highest	1.25M	7.6e-6	4.98

Why pair-bootstrap is misaligned here. A naive bootstrap that re-samples (q, d) pairs estimates a different quantity than the conditional on-counts efficiency we care about, for three reasons. (i) Averaging over $\sim 60\%$ near-singleton pairs under uniform produces a smooth resampling distribution whose tightness reflects averaging-of-noise rather than estimator precision. (ii) Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, so concentrated weight schemes are systematically penalised by the resampling design, not by their statistical behaviour. (iii) Pair-bootstrap entangles estimator efficiency with model fit; $\tilde{V}$ separates them (Section 4). The appropriate empirical baseline against which to validate the plug-in is the click-resample (parametric Bernoulli bootstrap conditional on counts), which we adopt for calibration.

Calibration. Empirical click-resample SE and the plug-in $\tilde{V}$ agree closely on KDD: at $f = 0.001$ within 1% (uniform 0.1306 vs. 0.1321; harmonic 0.1113 vs. 0.1120); at $f = 0.01$ within 2% (uniform 0.0387 vs. 0.0379; harmonic 0.0285 vs. 0.0280). The plug-in is well calibrated in this diagnostic.

4 Real-Log Results: KDD Cup 2012 Track 2

Data. Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique $(q, \text{ad}, \text{pos})$ cells, 8.2M clicks (CTR 3.49%). Pair (1, 2) interventional set: 4.61M (q, ad) pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \leq 5$.

Headline (Table 1). $\text{ARE}(\text{harmonic}, \text{uniform}) = 4.53$. **Uniform requires $\sim 4.5\times$ the impressions of harmonic for matching precision**, with $\text{SE}_U / \text{SE}_H \approx 2.13$. The min heuristic [1] achieves 4.41, the count-only oracle 4.57; harmonic is the principled parameter-

Stratified ARE (Table 2). Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by 5× on the high-information quartile. The gain comes from *not* treating singletons as equal to million-impression cells.

Weight-cap robustness. The harmonic ESS of 1,849 out of 4.61M pairs invites a concern that the efficiency gain rests on a few extreme cells. Capping ω_i^h at the 99.99th percentile (threshold 2,079; 462 pairs capped) lifts ESS to 5.2×10^4 and still gives $ARE = 4.12$; at the 99th percentile, $ARE = 2.83$ (ESS 7.6×10^5); even at the 90th, $ARE = 1.86$. Harmonic beats uniform robustly across capping levels.

Estimand under PBM mis-specification. PBM is a known approximation; on real logs different weight schemes estimate different functionals of cell-specific ratios. Specifically, the population limit of $\hat{R}$ under any weight ω is

$$R_\omega = \frac{\sum_i \omega_i \theta_k^i}{\sum_i \omega_i \theta_{k'}^i},$$

which equals $p_k/p_{k'}$ only when $\theta_k^i/\theta_{k'}^i$ is the same constant across cells—i.e. when PBM holds exactly. Under mis-specification each scheme thus targets a different weighted average of the cell-specific ratios $\rho_i = \theta_k^i/\theta_{k'}^i$, (uniform $\rightarrow 1.68$, harmonic $\rightarrow 1.71$, θ -aware oracle $\rightarrow 2.34$ on KDD). The plug-in $\tilde{V}$ comparison is a precision statement at fixed ω —a model-based efficiency comparison—and does *not* claim that one R_ω is the “correct” position-bias ratio. Choosing ω thus involves a precision/estimand trade-off; harmonic’s attractive property is that, among count-only schemes, it concentrates on high-information cells (where ρ_i is estimated most precisely) without depending on noisy plug-in $\hat{\theta}_i$.

Quantifying PBM mis-specification (Cochran’s Q). PBM is a known approximation, not a literal model – prior work [4, 12, 13] has long established that real logs violate it. The relevant question is how *large* the deviation is. Cochran’s Q on the 16,132 high-information pairs (≥ 5 clicks at each of positions 1 and 2) gives $Q = 6.17 \times 10^4$, $df = 16,131$, $Q/df = 3.83$: per-cell odds-ratio variability is $\sim 3.8 \times$ what binomial sampling alone would produce, well outside Poisson dispersion. This magnitude is the quantitative driver of the estimand drift discussed in Paragraph 4; plug-in $\tilde{V}$ keeps the efficiency comparison well-posed by working at fixed ω .

5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

Anchor-item imbalance sweep (synthetic). Following [1], $M=10$ positions, $p_k = k^{-1}$, anchor-item model with traffic split s and imbalance ratio r , 500 trials. At $s = 0.80$, $r = 100$ harmonic reduces variance by 43.6% vs. uniform, 5% vs. min; reduction grows to $\sim 54\%$ at $s = 0.90$, $r = 50$. Across the (s, r) grid, the data-equivalence factor $DEF = b_{\text{uni}}/b_{\text{harm}}$ has median 1.31 and maximum 1.94.

Yahoo-LTR semi-synthetic. Same simulator but with real graded relevance from Yahoo Learning to Rank Challenge Set 1 (6936 test queries; clicks simulated via PBM). Harmonic reduces variance by $\sim 40\%$ vs. uniform at imbalance $r = 50$ (Var 0.00015 $\rightarrow$ 0.00009; $\sim 458 \times$ lower MSE than aggressive clipping). Production-style traffic simulation with $K=5$ concurrent rankers (one default at 80% + four

treatment at 5% each, swap probability 0.5): under uniform traffic only 0.4% of pairs exceed ratio 10, but under production traffic 16.7% do; harmonic’s MSE reduction grows from 6.3% to **17.2%**, confirming the gain scales with traffic-induced imbalance.

6 Conclusion

Framing harmonic weighting as the count-only inverse-variance optimum (Theorems 2–4) and replacing bootstrap with the plug-in delta-method variance $\tilde{V}$ (Eq. 2) yields rigorous, calibration-verified efficiency comparisons on heavy-tailed real click logs. On the full KDD Cup 2012 Track 2 log, $ARE(\text{harmonic, uniform}) = 4.53$, rising to $\sim 5 \times$ on information-bearing cells, with PBM mis-specification quantified separately by Cochran- Q ($Q/df = 3.83$). Future work: shrinkage variants of the θ -aware oracle (Theorem 4 with $\hat{\theta}_i$ regularised); click models beyond PBM [4, 12, 13] where the intervention-harvesting family must be reformulated.

GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 2–4 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review. Full code, scripts, and data-processing pipelines are provided as supplementary material.

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